Week 10
Practice Assignment
Mathematics for Data Science - 2

1 Multiple Choice Questions (MSQ)

1. Match the functions of two variables in Column A with their directions in which the directional derivatives at (0,0) is maximized given in Column B.

	Functions of two variables		Direction of maximum directional derivative at (0,0)
(1)	$f(x,y) = x^2 - xy + y^2$	(i)	(0,-1)
(2)	$f(x,y) = 3x^2 + 3x + y$	(ii)	(1,-3)
(3)	$f(x,y) = e^{x+y} - 4y$	(iii)	(0,0)
(4)	$f(x,y) = sin(y) + cos(x)$	(iv)	(3,1)
		(v)	(0,1)
		(vi)	(1,3)

Table: M2PA11.1

- ◯ Option 1: (1) $\longrightarrow$ (v)
- ◯ **Option 2:** (1) $\longrightarrow$ (iii)
- ◯ **Option 3:** (2) $\longrightarrow$ (iv)
- ◯ Option 4: (3) $\longrightarrow$ (iv)
- ◯ Option 5:(2) $\longrightarrow$ (vi)
- ◯ **Option 6:** (4) $\longrightarrow$ (v)
- ◯ Option 7: (3) $\longrightarrow$ (vi)
- ◯ Option 8: (4) $\longrightarrow$ (i)

<u>Solution</u>: Maximum directional derivative is along the direction of ∇f.

1) $f(x,y) = x^2 - xy + y^2$

$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right) = (2x - y, 2y - x)$

$\nabla f(0,0) = (0,0)$.

2) $f(x,y) = 3x^2 + 3x + y$

$\nabla f = (6x+3,\ 1) \Rightarrow \nabla f(0,0) = (3,1).$

3) $f(x,y) = e^{x+y} - 4y$

$\nabla f = (e^{x+y},\ e^{x+y} - 4) \Rightarrow \nabla f(0,0) = (1,-3).$

4) $f(x,y) = \sin y + \cos x$

$\nabla f = (-\sin x,\ \cos y) \Rightarrow \nabla f(0,0) = (0,1).$

2. Which of the following statements is(are) correct?

- ○ Option 1: Maximum directional derivative at (0,0) for the function $f(x, y) = |x| + |y|$ is 1.
- ○ **Option 2:** The graph of the linear approximation to a function $f(x, y)$ at a given point P is the tangent plane to $f(x, y)$ at the point P.
- ○ Option 3: A tangent plane to a function $f(x, y)$ exists at every point.
- ○ **Option 4:** All directional derivatives of a function $f(x, y)$ at a point P are 0 if the tangent plane is parallel to the XY-plane.

<u>Option 1</u>: The directional derivatives does not exist at (0,0).
Infact, f_x, f_y do not exist at (0,0).

$$f_x(0,0) = \lim_{h \to 0} \frac{f((0,0)+h(1,0)) - f(0,0)}{h}$$

$$= \lim_{h \to 0} \frac{f(h,0)}{h} = \lim_{h \to 0} \frac{|h|}{h} \to \text{limit does not exist.}$$

<u>Option 2</u>: This is the geomertical interpretation of tangent Plane.

<u>Option 3</u>: Not true. <u>Ex:</u> $f(x,y) = |x| + |y|$ at Point (0,0).

<u>Option 4</u>: Tangent Plane Parralled to xy-Plane means both f_x and $f_y = 0$.

Therefore the directional derivative at $P = (a,b)$ in the direction of a unit vector "u" is

$$D_u f(a,b) = \nabla f(a,b) \cdot u = (f_x(a,b), f_y(a,b)) \cdot u$$

$$= (0,0) \cdot u = 0$$

3. Which of the following is(are) critical points of the scalar valued multivariable function $f(x, y) = 6x^2 - 2x^3 + 3y^2 - 6xy$?

- ◯ **Option 1:** (1,1)
- ◯ **Option 2:** (0,0)
- ◯ Option 3: (1,-1)
- ◯ Option 4: No critical points.

$f(x,y)$ is a Polynomial function. Therefore both f_x and f_y exist for all $(a,b) \in \mathbb{R}^2$.

$\Rightarrow (a,b)$ is a Critical Point Iff $f_x(a,b) = f_y(a,b) = 0$.

$f_x = 12x - 6x^2 - 6y$ — (1)

$f_y = 6y - 6x$ — (2)

$f_y = 0 \Rightarrow 6y - 6x = 0 \Rightarrow y = x$

Substitute $y = x$ in eq. (1), we get

$$12x - 6x^2 - 6x = 0$$

$$\Rightarrow 6x - 6x^2 = 0 \Rightarrow x = 0, 1$$

If $x = 0 \Rightarrow y = x = 0$ & $x = 1 \Rightarrow y = x = 1$.

The Critical Points are $(0,0)$ and $(1,1)$.

2 Numerical Answer Type (NAT)

4. Consider two functions f_1, f_2 defined as

$$f_1(x, y, z) = \ln(x^2 + y^2 + 2z^2)$$

and

$$f_2(x, y, z) = \tan^{-1}(x^2 + y^2 + 2z^2).$$

If $L_{f_1}(x, y, z)$ and $L_{f_2}(x, y, z)$ are the linear approximation of $f_1(x, y, z)$ and $f_2(x, y, z)$ at point (1, -1, 1), then find the value of $17\,(\nabla L_{f_1}(x, y, z) \cdot \nabla L_{f_2}(x, y, z))$.

- $f_1(x,y,z) = \ln(x^2+y^2+2z^2)$

$$\frac{\partial f_1}{\partial x} = \frac{1}{x^2+y^2+2z^2} \cdot 2x \Rightarrow \frac{\partial f_1}{\partial x}(1,-1,1) = \frac{2}{4} = \frac{1}{2}$$

$$\frac{\partial f_1}{\partial y} = \frac{1}{x^2+y^2+2z^2} \cdot 2y \Rightarrow \frac{\partial f_1}{\partial y}(1,-1,1) = \frac{-2}{4} = -\frac{1}{2}$$

$$\frac{\partial f_1}{\partial z} = \frac{1}{x^2+y^2+2z^2} \cdot 4z \Rightarrow \frac{\partial f_1}{\partial z}(1,-1,1) = 1.$$

$$L_{f_1}(x,y,z) = f_1(1,-1,1) + \nabla f_1(1,-1,1) \cdot (x-1, y+1, z-1)$$

$$= \ln(4) + \tfrac{1}{2}(x-1) - \tfrac{1}{2}(y+1) + z - 1$$

$$= \frac{x}{2} - \frac{y}{2} + z + \ln(4) - 2.$$

$$\nabla L_{f_1} = \left(\tfrac{1}{2}, -\tfrac{1}{2}, 1\right)$$

- $f_2(x,y,z) = \tan^{-1}(x^2+y^2+2z^2)$.

$$\frac{\partial f_2}{\partial x} = \frac{1}{1+(x^2+y^2+2z^2)^2} \cdot 2x \Rightarrow \frac{\partial f_2}{\partial x}(1,-1,1) = \frac{2}{17}.$$

$$\frac{\partial f_2}{\partial y} = \frac{1}{1+(x^2+y^2+2z^2)^2} \cdot 2y \Rightarrow \frac{\partial f_2}{\partial y}(1,-1,1) = \frac{-2}{17}.$$

$$\frac{\partial f_2}{\partial z} = \frac{1}{1+(x^2+y^2+2z^2)^2} \cdot 4 \Rightarrow \frac{\partial f_2}{\partial z}(1,-1,1) = \frac{4}{17}.$$

$$L_{f_2}(x,y,z) = f_2(1,-1,1) + \nabla f_2(1,-1,1) \cdot (x-1, y+1, z-1)$$

$$= \tan^{-1}(4) + \left(\frac{2}{17}, \frac{-2}{17}, \frac{4}{17}\right) \cdot (x-1, y+1, z-1)$$

$$= \frac{2}{17}x - \frac{2}{17}y + \frac{4}{17}z + \tan^{-1}4 - \frac{8}{17}$$

$$\nabla L_{f_2} = \left(\frac{2}{17}, \frac{-2}{17}, \frac{4}{17}\right)$$

$$\nabla f_1 \cdot \nabla f_2 = \left(\frac{1}{2}, -\frac{1}{2}, 1\right)\left(\frac{2}{17}, \frac{-2}{17}, \frac{4}{17}\right)$$

$$= \frac{1}{17} + \frac{1}{17} + \frac{4}{17} = \frac{6}{17}.$$

$$17(\nabla f_1 \cdot \nabla f_2) = 17 \cdot \frac{6}{17} = 6.$$

5. Suppose $z = f(x, y) = \frac{-7}{6}x^2 + \frac{1}{3}y^2 + \frac{11}{6}$ represents a surface. Let (x_0, y_0, z_0) be the point where the tangent plane to the given surface is parallel to the plane $7x - 8y + 3z + 2 = 0$. Find the value of $x_0 + 2y_0 + 3z_0$.

The tangent plane should be parallel to

$$7x - 8y + 3z + 2 = 0$$

That is, $-\frac{7}{3}x + \frac{8}{3}y - \frac{2}{3} = z.$

A plane parallel to this will be of the form

$-7/3\, x + 8/3\, y + K = z$ for some constant K.

$f_x(x_0, y_0) = -7/3 \Rightarrow -7/3\, x_0 = -7/3 \Rightarrow x_0 = 1$

$f_y(x_0, y_0) = 8/3 \Rightarrow 2/3\, y_0 = 8/3 \Rightarrow y_0 = 4$

$z_0 = f(x_0, y_0) = -7/6 + 16/3 + 11/6 = 6.$

$\Rightarrow x_0 + 2y_0 + 3z_0 = 1 + 8 + 18 = 27.$

6. Find the number of points where the tangent plane to the function $f(x, y) = x^3 + y^3 + 3x^2 - 9x - 12y + 20$ at the points is parallel to the XY-plane. [Answer: 4]

Equation of a Plane Parallel to the XY-Plane is $Z = C$, where C is a Constant.

If the equation of the tangent Plane at Point (x_0, y_0) is $Z = C$

Then $f_x = 0 = f_y$ at (x_0, y_0).

$$f(x,y) = x^3 + y^3 + 3x^2 - 9x - 12y + 20.$$

$$f_x = 3x^2 + 6x - 9 = 0 \Rightarrow x = -3, 1$$

$$f_y = 3y^2 - 12 = 0 \Rightarrow y = \pm 2.$$

There are 4 Points where the tangent Plane is Parallel to the XY-Plane. These Points are

$$(-3, 2), (-3, -2), (1, 2), (1, -2).$$

3 Comprehension Type Question:

• Suppose $ax + by + cz = d_1$ is a tangent plane to the function $f(x, y) = x^3 + 3y^3 + 9xy$ at (-1,2) and $ax + by + cz = d_2$ is a tangent plane to the same function $f(x, y)$ at $(\alpha, 1)$.

Use the above information to answer questions 7-8:

7. Which of the following options is(are) true?
   - ◯ Option 1: α= -2
   - ◯ **Option 2:** $\alpha = 2$
   - ◯ Option 3: There are only two possible candidates for α.
   - ◯ **Option 4:** $|d_2 - d_1| = 12$

$f(x,y) = x^3 + 3y^3 + 9xy.$

$f_x(x,y) = 3x^2 + 9y \Rightarrow f_x(-1,2) = 21.$

$f_y(x,y) = 9y^2 + 9x \Rightarrow f_y(-1,2) = 27.$

Equation of the tangent Plane at $(-1,2)$ is

$Z = f(-1,2) + (21, 27)\cdot(x+1, y-2)$

$= 5 + 21x + 21 + 27y - 54$

$Z = 21x + 27y - 28.$

$\Rightarrow 21x + 27y - Z = 28$

By Comparing the coefficients we get

$a = 21,\ b = 27,\ C = -1,\ d_1 = 28.$

It is given that $21x+27y-z=d_2$ is the equation of the tangent plane of $f(x,y)$ at point $(\alpha, 1)$.

$$\Rightarrow f_y(\alpha, 1) = 27.$$

$$\Rightarrow 9\alpha+9=27 \Rightarrow \boxed{\alpha=2.}$$

$\Rightarrow (2,1,\overset{\shortparallel}{f(2,1)})$ is a point of the plane $21x+27y-z=d_2$

$$\Rightarrow 42+27-29=d_2 \Rightarrow d_2=40$$

$$\Rightarrow |d_2-d_1|=|40-28|=12.$$

8. Find the value of $\frac{2a+b+c}{d_1+d_2}$. [ans: 1]

$$\frac{2a+b+c}{d_1+d_2} = \frac{42+27-1}{68} = 1.$$

- Let $L(x, y)$ be the linear approximation to $f(x, y) = xe^y + ln(x + y)$ at point (1,0).

Use the above information to answer questions 9-10:

9. Suppose $L(x, y) = ax + by + c$, where $a, b, c \in \mathbb{R}$. Find the value of $a + b - c$. [Ans: 5]

$$L(x,y) = f(1,0) + \nabla f(1,0) \cdot (x-1, y-0).$$

$$\frac{\partial f}{\partial x} = e^y + \frac{1}{x+y} \Rightarrow \frac{\partial f}{\partial x}(1,0) = 2.$$

$$\frac{\partial f}{\partial y} = xe^y + \frac{1}{x+y} \Rightarrow \frac{\partial f}{\partial y}(1,0) = 2.$$

$$L(x,y) = 1 + (2, 2) \cdot (x-1, y-0) = 2x + 2y - 1.$$

$$\Rightarrow a + b - c = 2 + 2 - (-1) = 5$$

10. Find the approximate value of the function $f(x, y)$ at (1.1,0.1) using the linear approximation function $L(x, y)$. [Answer: 1.4]

$$L(1.1, 0.1) = 2(1.1) + 2(0.1) - 1 = 1.4.$$